Module 6

Exercise 1 (Intro to NodeJs)

1. Installing node latest version and npm

A1.)

```
dhruv-pandey@TTNPL-dhruv:~ $# Download and install nvm:
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.4/install.sh | bash

# in lieu of restarting the shell
\. "$HOME/.nvm/nvm.sh"

# Download and install Node.js:
nvm install 24

# Verify the Node.js version:
node -v # Should print "v24.14.0".

# Verify npm version:
npm -v # Should print "11.9.0".
```

```
dhruv-pandey@TTNPL-dhruv:~ $node -v
v24.13.1
dhruv-pandey@TTNPL-dhruv:~ $npm -v
11.8.0
dhruv-pandey@TTNPL-dhruv:~ $
```

2. Read file asynchronously and synchronously

A2.) Code
Asynchronous

```
const fs=require('fs');


const read=()=>{
    console.log('Start');
    fs.readFile('./file.txt', 'utf-8', function(err, content) {
        if (err) {
            console.log(err);
            return;
        }
        console.log(content);
    });



    console.log('end')
}
```

Synchronous

```
const readSync=()=>{
    console.log('start');

    const content=fs.readFileSync('./file.txt','utf-8');
    console.log(content);

    console.log('end');
}

// console.log('Reading async\n');
// read()

console.log('Reading sync\n')
readSync();
```

Output
Asynchronous

```
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$node ReadingFile.js
Reading async

Start
end
Hi Developer
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$
```

Synchronous

```
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$node ReadingFile.js
Reading sync

start
Hi Developer
end
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$
```

3. Install dependencies and devdependencies

A3.) Dependencies

```
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$npm i express

added 65 packages, and audited 66 packages in 3s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$
```

Dev Dependencies

```
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$npm i -D nodemon

added 26 packages, and audited 92 packages in 2s

27 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$
```

Output

```
  "dependencies": {
    "express": "^5.2.1"
  },
  "devDependencies": {
    "nodemon": "^3.1.14"
  }
}
```

4. Creating module and import(require) & export

A4.) Module

Module6 > Exercise1 > module.js > <unknown>

```
function Add(a,b){
    return a+b;
}
function Sub(a,b){
    return Math.abs(a-b);
}

module.exports={Add,Sub}
```

Import

Module6 > Exercise1 > import.js > ...

```
const {Add,Sub}=require('./module');

const add=Add(2,5);
const sub=Sub(2,5);

console.log(`The Addition of 2,5 is ${add}\n The Subtraction of 2,5 is ${sub}`);
```

Output

```
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$node import.js
The Addition of 2,5 is 7
 The Subtraction of 2,5 is 3
dhruv-pandey@TTNPL-dhruv:Exercise1 (master)$
```

5. Creating an http server

A5.) Code

Module6 > Exercise1 > server.js > server > http.createServer() callback

```
const http=require('http');

const server=http.createServer((req,res)=>{
    res.end("Hello ");
})

server.listen(4000);
```

Output

localhost:4000

Hello